Threshold Gating and Information Disclosure in Observational Learning

Abstract

Early public actions can amplify noisy signals and generate inefficient information cascades in sequential financial decisions. This study examines whether threshold gating can improve collective accuracy under different disclosure regimes. We develop an agent-based computational model in which ten agents receive binary private signals with 0.60 accuracy and decide sequentially. Across 3,000 independent groups per treatment, we compare pure action observation (PAO), optional disclosure (OD), selective disclosure (SD), and incentivized optional disclosure (IOD). A dual-threshold mechanism delays the use of weak public signals and permits silence to carry directional information only after the public evidence becomes sufficiently strong. An LLM-based module generates heterogeneous information-processing weights and provides a small-sample consistency check. Without threshold gating, disclosure regimes that screen information or constrain low-quality disclosure perform best: mean accuracy over the last five decisions is 0.888 under SD and 0.856 under IOD, compared with 0.753 under PAO and 0.660 under OD. Introducing the baseline thresholds raises accuracy from 0.753 to 0.798 under PAO and from 0.660 to 0.697 under OD by limiting early reliance on noisy public histories. The effects are heterogeneous when disclosure already filters information: accuracy rises from 0.888 to 0.910 under SD but falls from 0.856 to 0.843 under IOD, indicating a cost from delaying useful signals. Results remain directionally stable across random seeds, and sensitivity tests identify an effective range of 2-3 for the public-information threshold and 2-4 for the silence threshold. The LLM-augmented simulations preserve the broad advantage of screened or incentivized disclosure while increasing behavioral variation. These findings show that threshold gating is a targeted corrective for weak information environments, not a universal addition to disclosure rules. Effective design therefore depends on matching information-use thresholds to the screening capacity of the underlying disclosure regime.

Keywords: Observational learning; Information disclosure; Threshold gating; Information cascades; Agent-based simulation; Large language models